Data Warehousing – Concise Overview

Definition

A data warehouse is a centralized repository designed to store large volumes of integrated historical data from multiple sources. Unlike operational databases, it is optimized for analysis, reporting, and decision support rather than daily transactions.

Main Characteristics

Data warehouses are subject-oriented, integrated, time-variant, and non-volatile. Subject-oriented means data is organized around major business subjects such as customers or sales. Integrated means data from various systems is cleaned and standardized. Time-variant refers to storing historical data. Non-volatile means data is mostly read and rarely modified.

Architecture

A typical architecture includes: data sources (ERP systems, applications, databases), an ETL process (Extract, Transform, Load), a staging area, the warehouse itself, and analytical tools such as dashboards or BI systems.

ETL Process

ETL extracts data from sources, transforms it by cleaning and standardizing formats, then loads it into the warehouse. ETL ensures high-quality and consistent data.

Schemas

Common modeling approaches include the Star Schema and Snowflake Schema. The Star Schema uses fact tables connected to dimension tables, while Snowflake normalizes dimensions into multiple related tables.

Benefits

Benefits include better business intelligence, faster reporting, historical trend analysis, improved decision-making, and a unified view of organizational data.

Applications

Data warehouses are widely used in healthcare, finance, retail, telecommunications, and business intelligence projects to support strategic decisions and analytics.